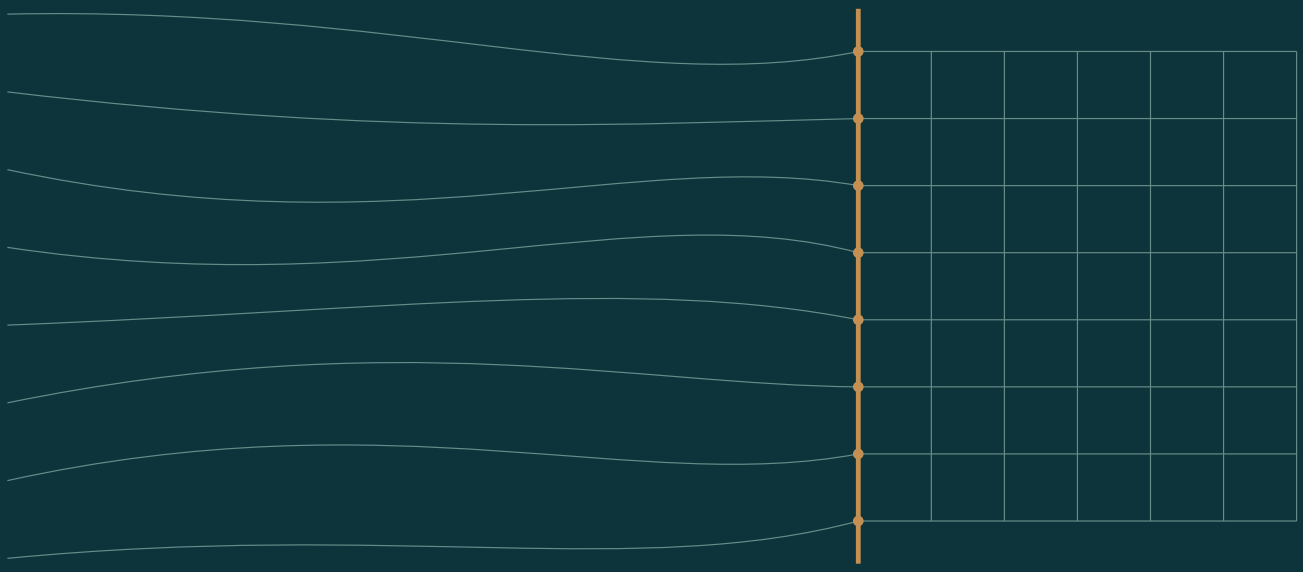
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A boundary hypothesis for genesis and structural stabilization

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ORIGIN / STRUCTURE / STABILIZATION



A SOURCE-FREE ORIGIN HYPOTHESIS WITH CONDITIONAL MATHEMATICAL MODELS

Complete argument, proofs, calculations, and reproducibility appendix

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Abstract

This paper proposes that existence has a genesis without antecedent actuality, and investigates how constraints established with actuality could stabilize its subsequent development. Absolute nothingness means the absence of all positive actuality, including physical laws, fields, spacetime, and producers. It is represented by a boundary claim, not an earlier physical state. The absence of antecedent restrictions motivates the origin hypothesis; the inference from absence to possible realization remains an explicit conjecture. Subsequent development is represented by positive structures whose properties determine admission constraints. A finite fermionic construction gives exact formulas for additions, closure times, and continuing internal dynamics after the addition channels close. A separate packing model shows how infinitely many candidate forms can coexist with finite occupancy, and establishes eventual closure under explicit accessibility assumptions, including biased, history-dependent exploration. Further results distinguish exact closure from operational stability and exhibit finite-start and beginningless histories with identical represented futures. The deductions, proofs, worked calculations, and a verification program are included. A conditional cosmological interface states how a physical version would have to connect structural development with spacetime, conserved stress-energy, and an expanding early universe. The models establish compatibility of the proposed post-genesis behaviors under stated assumptions. They do not establish inevitable realization, identify the first physical structure, or demonstrate that a Big Bang originated in this way.

1. The hypothesis and its intended explanation

The question is why anything exists, and how an answer might connect with the origin and subsequent lawfulness of a universe. The proposed answer is that actuality can originate without antecedent actuality, and that the structure thereby established supplies constraints on further actualization. The explanatory ambition is that all physical actuality ultimately has such an origin. The mathematical task developed here is to construct consistent positive histories exhibiting the proposed constraint feedback and stabilization, while keeping their origin assumption visible.

The proposal has three components. First, there is no antecedent actuality at the proposed genesis boundary. Second, realization establishes properties and relations, which constrain what can subsequently become actual. Third, further development can reach a regime with persistent governing conditions, while ordinary causal events continue. Stability can mean exact closure of specified addition channels or sufficiently small remaining effects over a specified observation period.

The mnemonic Nothing -> Everything -> Something records this proposal. Nothing denotes absolute absence. Everything denotes unrestricted nonactual possibility, rather than an actual collection of worlds or objects. Something denotes determinate actuality whose properties bear constraints. The first two expressions describe the proposed boundary from complementary perspectives. They do not name consecutive states in a preexisting time. Unrestricted means unrestricted by antecedent actual conditions; it does not assert that contradictory descriptions are jointly realizable or that every candidate must become actual. Once actuality has temporal relations, later additions and ordinary events can form actual sequences.

Two distinctions prevent changes of subject. A physical vacuum is already an actual structure with properties. It therefore cannot stand in for absolute nothingness. Also, the statement that all existence originates without antecedent actuality does not entail that every coherent description is an existing world. A plenitude of actual worlds is an additional hypothesis, unnecessary for the constructions here. Parfit discusses such maximality principles as possible explanations of existence [1]. This paper retains the source-free genesis question and the development of constraints as its central subjects.

The distinction between an origin without a cause and an origin caused by a thing called nothing is equally important. Hume explicitly separates the ideas of beginning to exist and having a cause [2]. Nothingness is not assigned productive agency here. The proposed absence is an enabling condition in the author's

hypothesis: no antecedent actual restriction must be overcome. Whether that condition is sufficient for realization is the principal unresolved origin question.

1.1 Terminology and scope

Actuality means instantiated positive content. A positive structure includes its constituents, properties, relations, and any response dispositions specified for it. Absolute nothingness denotes no such content at all. The phrase no antecedent actuality expresses the corresponding origin claim without placing an empty state before the first positive structure. It excludes a prior actual context or substrate, as well as an external producer or ground on which the structure depends for its actuality. A first event with no earlier event in a diagram does not by itself establish the absence of every such context or dependence.

Genesis means first actuality relative to this complete source-free interpretation. It need not mean a transition taking place in an external time. A first packet is a proposed positive whole, including any mutually required constituents and its initial response content. A direct addition is an enlargement admitted by that whole's represented structure. An ordinary event is an evolution within the represented actuality. These are distinctions in the model, whose physical interpretation must be supplied before they identify categories in cosmology.

Finite occupancy, finitely many direct additions, finite spatial volume, and finite duration are different propositions. None follows merely from calling something determinate. The finite examples below establish particular capacities and addition counts. An infinite physical universe, continuing ordinary events, or further causal domains remain questions outside those finite-capacity theorems.

2. The genesis boundary and the status of the origin argument

The origin argument can be stated without inserting a physical state of nothingness:

H1. Structural restriction. Operative restrictions borne by actual structure depend on instantiated properties or relations.

H2. Absolute boundary. At the proposed genesis there is no antecedent actuality and hence no restriction borne by antecedent actual structure.

H3. Boundary admission. At least one internally coherent positive structure can obtain with no antecedent actuality. The absence identified in H2 is proposed as the enabling condition for this admission.

H4. Structural development. Once positive structure obtains, its properties constrain later direct additions and ordinary evolution.

H1 and H2 jointly remove one class of obstacle to genesis. H3 states the substantive bridge from that absence to possible realization. It is not a consequence of H1 and H2 alone. Logical consistency, modal principles, or a proposed necessary dependence would require separate assessment. Equally, a universal requirement for an earlier producer is not assumed: extending familiar within-universe production requirements to genesis would itself need justification.

The thesis seeks an explanation, rather than merely a label for an unexplained first event. It proposes that absence of antecedent restrictions has explanatory significance. H3 is a claim of possible source-free realization; the explanatory thesis applies that claim to the origin of actual existence. The stronger assertion that realization is inevitable would additionally need a reason why non-realization requires a constraint, or some other argument excluding total absence. None of these steps is supplied by assigning a probability to a physical vacuum or by counting possible descriptions. No probability measure on nonactual possibilities is assumed here.

An internally consistent representation supports a conditional mathematical conclusion. A further interpretation identifies what its objects would be physically. A further philosophical argument would justify the possibility of that whole obtaining without antecedent actuality. These are distinct obligations. Chalmers's analysis of conceivability and possibility illustrates why a move between the first and third requires care [3]. This paper does not infer physical or metaphysical realization merely from the existence of symbols.

2.1 What a rooted representation establishes

Let an event domain E have a strict causal order, and let g be a minimal positive packet. A history can have g and later events without containing an earlier empty event. A clock, if used, belongs to the positive packet and its development. Mathematical parameters specifying such a history describe its positive content; they are not an actual law operating before actuality.

The interpretation that g has no external producer or ground is an explicit boundary hypothesis. Minimality in the causal order alone supplies only the weaker fact that there is no represented causal predecessor. A complete origin account must include every actual dependence it claims absent. It must also justify why the proposed positive content can obtain together, rather than removing an essential constituent to create apparent independence.

Proposition 1. Co-presence does not entail causal precedence. Requiring two positive constituents to obtain together does not logically require one to causally precede the other.

Proof. Consider a strict-order domain with minimal elements g and h and a later sequence e_1, e_2, ... Both g and h precede every e_i, and neither precedes the other. Require g and h to occur together in every represented alternative. Their co-presence holds throughout the class, but neither has a causal predecessor in it. This proves the stated distinction within the causal representation. It establishes neither absence of noncausal grounds nor concrete realizability of the represented alternatives.

More generally, a nonempty well-founded dependence relation has a minimal member, by the definition of well-foundedness. Existing actuality alone does not establish that the complete dependence relation is well-founded, uniquely rooted, or exhausted by temporal causation. No proof of a unique absolute beginning is inferred from that conditional observation.

2.2 The stronger self-collapse conjecture

The motivating intuition is that nothing enforces continued non-realization when nothing is actual. Its strongest interpretation is that total absence cannot obtain as a complete alternative. That is stronger than H3. It would require a reason that absence of restrictions entails some realization, rather than merely failing to prohibit it.

Stability and instability ordinarily describe states relative to a response law. Since no physical state or response law is assigned to absence, self-collapse cannot be introduced as an ordinary dynamical instability of nothingness. It can be retained as a name for the stronger conjecture just stated. The models below concern the development of actuality and provide no duration, decay rate, or probability of collapse for absence itself.

3. A common architecture for structural development

Let C be a class of positive structures X. Let $\theta(X)$ record their declared relevant properties and relations, and let $L(X)$ record their response constraints in common units and conventions. The structural hypothesis is:

$$L(X) = \Psi(\theta(X)). \tag{1}$$

Proposition 2. Property sufficiency. A map Ψ satisfying (1) exists on θ(C) exactly when equal declared profiles have equal responses throughout C.

Proof. If Ψ exists, equal inputs give equal outputs. Conversely, for every represented profile p define Ψ(p) as L(X) for a structure with θ(X)=p. The equal-response condition makes this definition independent of the representative. No dynamical response has been derived until θ and the corresponding response content have been independently specified.

Two structures with equal declared profiles and different responses refute the sufficiency of that profile. Adding the entire desired response to θ repairs the formal equation by definition, while leaving the response's origin unexplained. This distinction matters when the framework is applied to laws: consistency of a property-to-response map is not a derivation of the particular laws of our universe.

For a declared family H of candidate additions, let A(X,h) mean that h is admitted, and let $F_h(X)$ be the enlarged structure. Ordinary evolution is denoted D_t. Where an actual time and stochastic response are specified, take a nonnegative intensity q(X,h) that vanishes for inadmissible additions, and a measure ν on the mathematical labels. Require the integral to be finite wherever it specifies a total event rate:

$$U(X) = \{h \in H : A(X, h)\}, \qquad \lambda_{\text{add}}(X) = \int_H q(X, h)\, d\nu(h). \tag{2}$$

The measure and rate are post-genesis response assumptions, not a distribution over absolute nothingness. A candidate label represents a possible addition; it does not posit an actual reservoir of unrealized entities. For a countable family, λ_{add} is a sum. For a continuous family, zero total intensity need not mean that the descriptive set U is empty, since sets of zero measure can remain.

Exact structural closure means U(X) is empty for the declared family and ordinary evolution preserves that condition. Stochastic inactivity means λ_{add} is zero, a potentially weaker condition. Operational stability means that remaining observable effects obey a specified tolerance over a specified interval. These meanings are separated throughout.

Constraints can depend on occupation, arrangement, or other properties. They can tighten, weaken, or reorganize after an addition. A model with a fixed response rule and changing admitted channels illustrates structural feedback. Deriving the creation or evolution of the response rule itself is a further task. Whole-history formulations of laws also show that restrictions need not take the form of an earlier local enforcing force [4].

3.1 Determinacy and causation

A determinate structure excludes combinations inconsistent with its specified properties. It does not follow that every event in every coherent structure has an earlier producer. A complete description can contain an ordered history and an event declared to have no incoming causal relation without logical contradiction. A universal causal requirement would be an additional premise.

The narrower structural claim is sufficient here: if the actual properties of a particular structure exclude a given spontaneous addition, that addition is inadmissible within the model. This allows an origin hypothesis with no antecedent actuality to coexist with later restrictions. It does not infer the restrictions of our universe from bare determinacy, or infer their absolute validity from the absence of observed spontaneous macroscopic objects.

4. A finite fermionic benchmark

The first worked construction hypothesizes a finite mode structure, an internal degree of freedom, an initial positive occupation, and a response law as content of first actuality. It is a compatibility example. Its use of fermionic algebra is a mathematical choice, not an identification of the first physical constituents of the universe.

Choose an integer $M \geq 1$, an internal dimension $d \geq 1$, and an initial number $1 \leq n_0 \leq M$. Let $F(\mathbb{C}^M)$ be the finite fermionic Fock space. The represented space and addition operators are:

$$\mathcal{H} = \mathcal{F}(\mathbb{C}^M) \otimes \mathbb{C}^d, \qquad C_j = a_j^\dagger \otimes I_d, \qquad N = N_F \otimes I_d. \tag{3}$$

The annihilation and creation operators satisfy the canonical anticommutation relations $\{a_j, a_k^\dagger\} = \delta_{jk} I$ and $\{a_j, a_k\} = 0$. Here N_F is the sum of $a_j^\dagger a_j$. An occupation-zero vector would still belong to this already structured space. It is therefore not a representation of absolute nothingness.

The inventory of positive assumptions includes M, the mode algebra, the internal factor, the initial state, the clock used after genesis, the ordinary response K, the addition family, and the gain rule below. There is no loss channel in this benchmark. Ordinary evolution preserves total occupation, although it may redistribute occupation among modes. The model is not a derivation of these items from absence.

Proposition 3. Exclusion and finite channel closure. For every normalized state ψ in number sector n:

$$\sum_{j=1}^{M} C_j^\dagger C_j = MI - N, \qquad \sum_{j=1}^{M} \|C_j \psi\|^2 = M - n. \tag{4}$$

Proof. The anticommutation relation gives $a_j a_j^\dagger = I - a_j^\dagger a_j$. Sum over j and tensor with the internal identity to obtain the operator equality. Its expectation in ψ gives the norm equality. At n=M every nonnegative norm is zero, so every C_j annihilates the state. At n<M at least one norm is positive. Linear combinations of the C_j also annihilate the full sector.

The addition restriction changes with the actual state: the total available norm decreases as occupancy increases. The final closure concerns exactly the declared creation family. It establishes neither closure to every conceivable kind of actuality nor finite spatial extent.

4.1 Continuing activity in the closed sector

Let K be a Hermitian ordinary-response generator with [K,N]=0. Then $U_t = \exp(-itK)$ preserves every number sector and hence preserves closure at n=M. The full fermionic sector is one-dimensional, but its tensor product with the internal factor has dimension d. For d=2, a separable example is:

$$K = I_F \otimes \frac{\omega \sigma_z}{2}, \qquad \langle I_F \otimes \sigma_x \rangle_t = \cos(\omega t), \qquad \omega > 0. \tag{5}$$

Here σ_z has diagonal entries 1 and -1, and σ_x has off-diagonal entries 1 and diagonal entries zero. The initial internal state is $(|0\rangle + |1\rangle)/\sqrt{2}$. The unitary multiplies its components by $\exp(-i\omega t/2)$ and $\exp(i\omega t/2)$, so its x-expectation is $\cos(\omega t)$. Occupation stays full while the expectation changes. Thus closure of the addition family and distinguishable ordinary activity coexist. K has units of inverse time; a physical Hamiltonian, if assigned, is ħK.

4.2 What an unstructured response cannot supply

The following standard finite-dimensional observations identify additional content needed by a first packet.

Proposition 4. Covariant state-only Hamiltonian selection is stationary. If a Hermitian response $K(\rho)$, chosen from a density matrix alone, obeys $K(U\rho U^\dagger)=UK(\rho)U^\dagger$ for every unitary U, then $[K(\rho),\rho]=0$. Its instantaneous unitary equation leaves ρ stationary.

Proof. Every unitary acting separately within eigenspaces of ρ fixes ρ. Covariance requires it to fix $K(\rho)$. Independent phases remove matrix blocks connecting different eigenspaces. Arbitrary unitaries within an eigenspace force the corresponding block to be scalar. Therefore $K(\rho)$ commutes with ρ, including in degenerate cases, and $-i[K(\rho),\rho]=0$. An unmarked full matrix algebra likewise selects only a scalar invariant Hermitian generator, which gives only a global phase.

The result assumes full unitary covariance and a closed unitary response. A separately specified K, a smaller symmetry group, or a nonunitary response changes the premises. It limits a particular attempt to derive motion from bare state information; it does not prohibit emergence in general.

5. Addition dynamics and exact stabilization

Choose a positive inverse-time scale κ after genesis. In a normalized n-sector state, assign the intensities and normalized updates:

$$q_j(\psi) = \kappa \|C_j\psi\|^2, \qquad \lambda_n = \kappa(M-n), \qquad \psi \longmapsto \frac{C_j\psi}{\|C_j\psi\|}. \tag{6}$$

The total rate follows from (4); the proportionality κ, exponential waiting rule, and clock are stipulated response content. An admitted jump increases n by one. Between jumps, ordinary number-preserving response acts. The normalized update is used only when its channel has positive rate. Existing properties determine the admitted channels, while the assumed gain law determines their timing. An admissible higher sector need not become occupied if its rate is instead zero.

The number process has transition n to n+1 at rate $\kappa(M-n)$. Write $v=M-n_0$ for the initial vacancy count. This count law can be represented by v independent exponential filling times with rate κ. The representation does not assert independence of all quantum degrees of freedom.

Proposition 5. Count distribution and closure time. For elapsed actual time $t\geq 0$ and $0\leq j\leq v$:

$$\Pr[N(t) = n_0 + j] = \binom{v}{j}(1-e^{-\kappa t})^j e^{-\kappa t(v-j)}. \tag{7}$$

$$\mathbb{E}[N(t)] = M - ve^{-\kappa t}, \qquad \operatorname{Var} N(t) = v(1-e^{-\kappa t})e^{-\kappa t}. \tag{8}$$

$$\Pr[T_{\text{close}} \leq t] = (1-e^{-\kappa t})^v, \qquad \mathbb{E}[T_{\text{close}}] = \frac{H_v}{\kappa}, \qquad H_v = \sum_{k=1}^{v} \frac{1}{k}. \tag{9}$$

For $v=0$, closure is present initially, the closure time is zero, and $H_0=0$; the cumulative distribution is one for $t\geq 0$.

Proof. In the auxiliary independent-vacancy representation, each vacancy has filled by t with probability $1-\exp(-\kappa t)$. The number filled has the binomial law (7), giving (8). Equivalently, substituting (7) into the pure-birth forward equations verifies it directly. Closure occurs when all v filling times have elapsed, giving the product in (9). Successive count-sector waits have means $1/(\kappa v)$, $1/(\kappa(v-1))$, ..., $1/\kappa$. The sum of their means gives H_v/κ, so the total closure time is finite almost surely. For $v>0$ every finite t nevertheless has a positive unsaturated ensemble tail.

Probability one is conditional on this stochastic response. It is neither logical inevitability of genesis nor a probability that the origin hypothesis is true. The theorem begins with positive actuality and makes no assignment of a waiting time in nothingness.

5.1 A quantum realization and its physical cost

The finite benchmark has a completely positive, trace-preserving Markovian representation of standard Gorini-Kossakowski-Sudarshan-Lindblad form [5,6]:

$$\dot{\rho} = -i[K,\rho] + \kappa \sum_{j=1}^{M} \left(C_j \rho C_j^{\dagger} - \tfrac{1}{2}\{C_j^{\dagger} C_j, \rho\} \right). \tag{10}$$

The jump representation in (6) is consistent with (10). Before normalization, its no-jump generator is -iK-(κ/2)(MI-N). On a fixed number sector the second term is scalar, so normalization leaves precisely the unitary response generated by K. Its squared norm decays at rate κ(M-n), giving the stipulated waiting law.

The adjoint generator applied to N is κ(MI-N), since $[N,C_j]=C_j$ and [K,N]=0. Taking the trace against ρ gives the mean equation in (8). If ρ is supported in the full sector, every dissipative term vanishes because both C_j and $C_j^{\dagger}C_j$ annihilate that sector. The ordinary internal response remains. Appendix A gives explicit operator details.

Invariant and attractive quantum subsystems are established subjects [7,8]. The construction uses those mechanisms as a benchmark. An effective physical gain equation usually describes a system coupled to other actual degrees of freedom. Such a reservoir, if used after genesis, belongs within the account of actuality. Effective occupation gain does not establish creation of new fundamental substance. Alternatively, (10) can be posited as a fundamental irreversible response of first actuality, but that interpretation requires independent physical motivation and conservation analysis.

Proposition 6. Finite autonomous unitarity does not yield irreversible absorption. A finite-dimensional autonomous unitary evolution cannot move positive probability from outside a permanently invariant subspace Z into Z.

Proof. If U_t Z is contained in Z, invertibility and equal finite dimension give U_t Z=Z. The orthogonal complement is also invariant, so the projector P_Z commutes with U_t. Consequently $\mathrm{tr}(P_Z\, U_{t\rho U t}^{\dagger})=\mathrm{tr}(P_{Z\rho})$. The weight in Z is constant. This proves the claim for the specified finite autonomous unitary setting.

Thus the irreversible gain law is substantive extra content. Open dynamics, time-dependent controls, larger state spaces, and approximate stability require their own analyses. Equation (10) is not derived from the closed unitary part of the benchmark.

5.2 A worked example and a rival

For M=4 and n_0=1, successive total rates are 3κ, 2κ, κ, and zero. Mean waits are 1/(3κ), 1/(2κ), and 1/κ. Hence the mean closure time is 11/(6κ), and the closure cumulative distribution is $(1-\exp(-\kappa t))^3$. The internal response (5) can continue after closure.

A rival uses $q_j=\gamma\|C_j\psi\|^2/(M-n)$ for n<M and zero at n=M, with γ>0. Its total rate stays γ until closure. It shares the algebra and the conditional choice of addition channel with (6), while producing different wait statistics. For n≤M-2, the ratio of the mean wait at n+1 to that at n is (M-n)/(M-n-1) in (6); in the rival it is one. Matching the initial rate does not match subsequent ratios. Algebra therefore determines some admission restrictions without uniquely determining the response times.

6. Infinitely many candidate forms and finite occupancy

The second construction illustrates how an infinite candidate family can have finite compatible occupancy. After genesis, assign forms attributes in [0,1] and require pairwise separation strictly greater than a fixed δ, where $0<\delta\leq 1$. The range and exclusion scale are actual properties stipulated for this example. A form's attribute is a model coordinate, not an already existing possible entity.

Proposition 7. Packing capacity and insertion volume. The maximum occupancy is M=ceil(1/δ). For ordered compatible n-form configurations Ω_n, their coordinate volume and mean admitted fraction under uniform configuration sampling are:

$$V_n = \frac{[1-(n-1)\delta]_+^n}{n!}, \qquad a_n = \frac{(n+1)V_{n+1}}{V_n} = \frac{[1-n\delta]_+^{n+1}}{[1-(n-1)\delta]^n}. \tag{11}$$

Here $[x]_+=\max(x,0)$, and the ratio is used only for valid occupancies with $V_n>0$. It is positive below M and zero at M.

Proof. For $n\geq 2$, ordered attributes $x_1<\ldots<x_n$ require $x_n-x_1>(n-1)\delta$, so $(n-1)\delta<1$. Occupancy one is always possible. The largest possible integer n is ceil(1/δ), and that value can be attained because $(n-1)\delta<1$ leaves room for strictly larger gaps. Translate $y_i=x_i-(i-1)\delta$. Up to boundary sets of zero volume, the y_i are ordered points in an interval of length $1-(n-1)\delta$. The ordered simplex has volume that length to the power n divided by n!, giving V_n. Each compatible (n+1)-configuration can be decomposed into an n-configuration and an inserted point in exactly n+1 ways. Therefore the integral over Ω_n of the admitted interval length is $(n+1)V_{n+1}$. Divide by V_n to obtain a_n.

The candidate interval contains uncountably many labels while occupancy is bounded. This is a precise version of one part of the infinite-to-finite intuition. Finitude follows from a bounded attribute range and a positive exclusion scale. The proof neither derives those properties from nothingness nor converts an actual infinite collection into a finite one.

6.1 Robustness under biased, history-dependent exploration

At each continuing opportunity, ordinary rearrangement preserves compatibility and count. Conditional on all past history, suppose its configuration measure dominates η times the uniform measure on Ω_n, with $0<\eta\leq 1$. Conditional on the history and that configuration, suppose the candidate density on [0,1] is at least ζ, with $0<\zeta\leq 1$. The remaining choices may depend on history and be biased. These are accessibility assumptions.

Proposition 8. Eventual persistent closure. Starting at any compatible positive count n_0, maximum occupancy is reached almost surely. If J counts further opportunities, then:

$$\mathbb{E}[J] \leq \sum_{n=n_0}^{M-1} \frac{1}{\eta\zeta a_n}. \tag{12}$$

Proof. At count n<M, the candidate floor and configuration domination give conditional success probability at least $c_n=\eta\zeta a_n>0$. After each failure the same bound holds conditional on the enlarged history. Iterated conditioning therefore bounds k consecutive failures by $(1-c_n)^k$. This tends to zero; summing the survival probabilities bounds the mean opportunities at n by $1/c_n$. Induct through the finite remaining counts and sum the bounds. At M, the capacity proof excludes further compatible additions. Ordinary compatibility-preserving, count-preserving changes cannot reopen that channel.

Independence between opportunities is unnecessary. If opportunities are also driven by an independent post-genesis Poisson clock of rate r>0, and choices use no knowledge of future clock increments, their mean time to closure is bounded by the right side of (12) divided by r. Indeed, the interval before

opportunity k has mean 1/r independently of whether closure has occurred by opportunity k-1; summing these contributions gives E[J]/r. A lower accessibility bound does not justify replacing the actual history-dependent process by an exact constant-rate process.

6.2 A calculated nonuniform family

Conditional on the past, draw the configuration and candidate independently with densities $[1+\xi(2\,\mathrm{mean}(S)-1)]/V_n$ on Ω_n and $1+\chi(2x-1)$ on [0,1], where $|\xi|<1$ and $|\chi|<1$. Here mean(S) is the arithmetic mean of the n attributes. Reflection about 1/2 normalizes both densities. Choose the biases before each opportunity's current draws. Their exact acceptance probability is:

$$a_n(\xi,\chi) = a_n(1-\xi\chi B_n), \qquad B_n = \frac{2\delta-(n-2)\delta^2}{3}, \qquad n < M. \tag{13}$$

Appendix B derives the formula. Uniform bounds on the absolute biases strictly below one supply accessibility floors and allow adaptive bias choices without destroying eventual saturation.

For $\delta=3/10$, $\xi=1/2$, and $\chi=3/4$, capacity is four. Acceptance probabilities at counts one, two, three, and four are respectively 35819/80000, 148/1225, 749/512000, and zero. The sum of the first three reciprocals is approximately 694.089. With independent refresh draws at fixed biases this is the exact mean opportunity count from count one. It is also an upper bound for adaptive choices satisfying $|\xi|\le 1/2$ and $|\chi|\le 3/4$ at every opportunity: B_n is positive for each of the three counts, so their conditional acceptance probabilities cannot be smaller than the displayed values. No physical duration is inferred from this count.

6.3 Failure examples

At $\delta=0.3$, the two-form configuration {0.25,0.75} admits no additional point even though capacity is four. Frozen configurations can therefore remain closed below maximum occupancy. A candidate rule concentrated entirely in excluded intervals can also prevent progression. Geometric capacity alone does not establish eventual filling.

Even positive exploration at every opportunity is insufficient if it fades too fast. Starting from that frozen arrangement, let independent refresh probabilities be $\varepsilon_k=2^{-(k+1)}$ for $k\ge 1$, with no other movement. Their sum is 1/2. Since $\log(1-\varepsilon_k)\ge -2\varepsilon_k$, the probability of no refresh is the positive product of $1-\varepsilon_k$, at least $\exp(-1)$. A uniform accessibility floor is sufficient, but not necessary: a divergent sum of lower success bounds already makes the corresponding failure product vanish. Finite mean settling time requires a stronger tail condition than eventual success alone.

7. Operational stability when additions continue

Exact closure is one possible endpoint. A second is a regime where additions continue but their consequences are small on an observation scale. A rate alone cannot establish this. Effect size, amplification, and the interval of observation must also be bounded.

For a precise finite-horizon version, count additions in $(s,s+\Delta]$, where $\Delta>0$. Assume a predictable nonnegative counting intensity bounded by a deterministic integrable function $\lambda_{bar}(t)$. Predictability means the rate uses information available before the event; the intensity convention requires the expected count on the interval to equal the expected integral of that rate. Suppose each event has a mark whose contribution to an observable at $s+\Delta$ has norm at most bA_Δ. Here b bounds the immediate effect and A_Δ bounds all subsequent amplification over the interval. Also assume that the total observable difference from a comparison beginning in the same state at s, with no later additions, is bounded by the sum of those marked contributions. This last assumption excludes uncontrolled nonlinear amplification.

Proposition 9. A residual-effect bound. Under these assumptions, for ε>0:

$$\mathbb{E}\|\Delta O\| \le bA_\Delta \int_s^{s+\Delta} \bar{\lambda}(t)\,dt, \qquad \Pr(\|\Delta O\| > \varepsilon) \le \frac{bA_\Delta}{\varepsilon} \int_s^{s+\Delta} \bar{\lambda}(t)\,dt. \tag{14}$$

Proof. The pathwise sum bound gives $\|\Delta O\| \le bA_\Delta$ times the number of events. By the defining compensator identity for a counting intensity, the mean event count is the mean integral of its intensity and is bounded by the integral of λ_{bar}. Taking expectations gives the first inequality. Markov's inequality gives the second. All needed effect and amplification bounds are assumptions about the actual positive system.

For example, if $\lambda_{bar}(t)=\lambda_0 \exp(-\alpha(t-s))$ with $\alpha>0$, the integrated rate is $\lambda_0(1-\exp(-\alpha\Delta))/\alpha$. This provides a finite-horizon stability criterion. It does not prove that the response of our universe has this form. A small constant rate can accumulate over long intervals; growth of A_Δ can erase an apparently small rate advantage. Negligible additions and exactly absent additions should therefore remain separate physical alternatives.

8. What later observations can establish about genesis

An origin hypothesis needs consequences that competing histories cannot reproduce. Identical positive packets with identical future response laws have identical conditional future observations, whether one is assigned a source-free interpretation or an earlier producer. Rates and closure statistics can test post-genesis response content while leaving that origin distinction unresolved.

8.1 A conditional restriction on persistent past response

Let $0 \le n_{min} < M$ be integers, and let N(t) be a time-inhomogeneous Markov pure-birth process on the integers from n_{min} to M. Assume transition rate $\kappa(t)(M-n)$ from n to n+1, no removals, and fixed capacity at every represented time. Let $\kappa(t) \ge 0$ be locally integrable, and write A(s,t) for its integral from s to t. Zero count in this system still presupposes the system's actual structure.

Proposition 10. Persistent accumulated response. For any initial count mixture at s:

$$\Pr[N(t) < M] \le 1 - (1 - e^{-A(s,t)})^{M - n_{\min}}. \tag{15}$$

If the process law is defined at all real times and A(s,t) diverges as s tends to minus infinity for every fixed t, then N(t)=M almost surely at every time.

Proof. From count n, each vacancy survives with probability exp(-A), giving unsaturation probability $1-(1-\exp(-A))^{M-n}$. This is largest at n_{min}, and mixing does not increase that maximum. If A diverges, the bound tends to zero for each fixed t. Apply the result simultaneously at rational times through a countable intersection. Monotonicity and the existence of rational times on both sides then imply saturation at all real times.

The conclusion restricts process laws satisfying those assumptions. It does not turn every exceptional zero-probability path into a metaphysical impossibility. It also does not infer absolute absence from the failure of a response to extend indefinitely into the past. Earlier actuality could have different capacity, removal channels, or response properties.

For constant $\kappa>0$ and an independently known ensemble deficit $d_0=M-E[N(t_0)]>0$, a duration T of unchanged response must satisfy:

$$T \le \frac{1}{\kappa} \log\left(\frac{M - n_{\min}}{d_0}\right). \tag{16}$$

Indeed the deficit T units earlier is $d_0 \exp(\kappa T)$ and cannot exceed $M-n_{min}$. This is a bound on response duration under an ensemble assumption, not an age inferred from a single observed system or evidence that nothing preceded that duration.

8.2 A finite-start and a beginningless history with the same future

Use an occupation-basis initial state and the separable internal generator (5), so ordinary evolution does not mix the occupation modes. Let $\kappa_0>0$ and $\tau>0$. History R begins with one fixed occupied baseline mode at $t=-\tau$ and has constant response κ_0 thereafter. History B contains that same occupied baseline mode throughout all real times, with response $\kappa_0 \exp(t/\tau)$ before zero and κ_0 after zero. Other modes may be filled in both histories. In B the response is positive at each finite time and tends to zero into the past.

Give each of the remaining M-1 vacancies a unit-exponential threshold E_i. A vacancy fills when its cumulative exposure reaches E_i. In R, exposure at zero is $\kappa_0\tau$. In B, the integral from minus infinity to zero is also $\kappa_0\tau$. For $t\geq0$ both exposures are $\kappa_0\tau+\kappa_0 t$.

Proposition 11. Matched future, different past. The histories can share exactly the same represented future despite differing in whether they have a first positive packet.

Proof. Use the same thresholds in both representations. Their occupied subsets then agree at every $t\geq0$. A vacancy with threshold below $\kappa_0\tau$ was filled before zero in both, but at generally different times. A threshold at or above $\kappa_0\tau$ produces the same filling time $(E_i-\kappa_0\tau)/\kappa_0$ in both. Give the internal system the same state at zero and the same subsequent generator; this is achieved by choosing its initial state in R by backward unitary evolution from zero, and defining its state in B by the same unitary group for all real times. The occupation projectors and internal states then coincide for $t\geq0$. Consequently the joint law of subsequent measurements of this represented system is the same. The model includes no independent clock-age variable or other record of the unequal pasts.

The exponential variables specify a probability law, not an actual reservoir before existence. In B, minus infinity is not a first instant. Each nonbaseline mode has a finite birth time almost surely, but the occupied baseline and its internal evolution have no first time. Thus a first addition need not be the beginning of all represented actuality. Allowing changing response properties restores a beginningless alternative that future observations in this benchmark cannot distinguish. The construction does not refute genesis; it identifies the additional past information required to infer it.

8.3 A specific loss of origin records

With the separable internal generator in (5), start in any definite singleton occupation and the same internal state. Fill every remaining mode once, in any order. The final fermionic vector is the unique full-occupation vector up to a sign. That sign is a global phase, so the final density matrix is identical. At equal elapsed times the internal state also agrees.

Consequently the final represented state alone cannot recover the original occupied mode or the order of additions. Timing records, environmental records, or internal couplings that store history could distinguish alternatives. Continuing activity after closure does not automatically preserve origin information.

9. A conditional interface with cosmology

The intended application is the emergence of a universe and the origin of its governing conditions. The models so far have occupation labels, abstract attributes, and chosen response laws. Connecting them to a Big Bang requires more than naming their rapid-growth phase. A physical version must identify spacetime or its precursor, an observable scale, stress-energy or its replacement, and a response whose large-scale limit can be compared with cosmology.

Classical sequential growth for causal sets supplies an existing example of mathematically specified causal development [9]. Such constructions demonstrate that discrete growth and causal restrictions can be studied precisely. Their existence does not establish H3 or make a reused growth law a new cosmological mechanism. A proposed model here would have to explain what its additions physically represent and how their constraints produce the needed continuum behavior.

9.1 An illustrative matching requirement

Suppose, as an additional physical hypothesis, that a regime after structural stabilization admits a spatially flat homogeneous and isotropic spacetime described by general relativity, constant G>0, and constant Λ. Let a(t)>0 be the scale factor and let dots denote cosmic-time derivatives. In units c=1 its conventional matching equations include [10]:

$$H^2 = \frac{8\pi G}{3}\rho_{\mathrm{tot}} + \frac{\Lambda}{3}, \qquad H = \frac{\dot{a}}{a}, \qquad \dot{\rho}_{\mathrm{tot}} + 3H(\rho_{\mathrm{tot}} + p_{\mathrm{tot}}) = 0. \tag{17}$$

Here ρ_{tot} and p_{tot} denote the energy density and pressure of the included material and structural sectors; they exclude the cosmological-constant contribution already written separately. These equations are a target for matching within their regime of validity. They are not assumed to operate before actuality or derived from the occupation model. If later additions are represented as homogeneous energy transfer Q between comoving perfect-fluid sectors g and m, a conservation-compatible split is:

$$\dot{\rho}_m + 3H(\rho_m + p_m) = Q, \qquad \dot{\rho}_g + 3H(\rho_g + p_g) = -Q. \tag{18}$$

The sectors in (18) are already actual parts of the proposed universe, with $\rho_{tot}=\rho_g+\rho_m$ and $p_{tot}=p_g+p_m$. Positive Q denotes transfer into m. Summing the equations recovers the total continuity equation. Treating Q as net creation of stress-energy with no compensating sector while retaining the same Einstein equations would require an additional consistent modification. The benchmark's κ and count n have no supplied conversion to Q, density, temperature, or the scale factor.

For illustration, take the ideal limit Q=0 and Λ=0 with positive density and total pressure p=ρ/3. The continuity equation gives $\rho=C\,a^{-4}$ for a constant C>0. On the expanding branch, the Friedmann equation then gives dot a=sqrt(8πGC/3)/a, and integration yields a proportional to $(t-t_*)^{1/2}$. If omitted contributions are only small, these relations are correspondingly approximate. This radiation-regime consequence follows from the added cosmological assumptions. It is not a derivation of radiation, spatial flatness, or the genesis boundary. Extrapolating t_* beyond that regime would need separate justification.

9.2 Observational and conceptual requirements

A cosmological version must recover the relevant expansion and thermal history, account for perturbations, and identify at least one consequence that sufficiently specified rivals do not share. Cosmic microwave background measurements constrain detailed cosmological models, as illustrated by the Planck parameter analysis [11]. The present paper supplies no fit to those data, no new primordial spectrum, and no derived value of a fundamental constant.

Quantum cosmological origin proposals also make mathematical commitments that should be compared explicitly. Hartle and Hawking specify a wave function on geometries and matter configurations [12]; Vilenkin proposes a tunneling creation model [13]. These accounts formulate origins within a quantum and geometric framework. Mathematical commitments alone do not imply an actual precursor, but neither do they establish this paper's inference from an absence of restrictions to possible realization. They provide comparisons for a physical origin proposal, rather than a justification for assigning a quantum state or tunneling probability to the absolute absence defined here. This paper does not identify a virtual particle, a metastable vacuum, or a quantum fluctuation with absolute nothingness.

The hypothesis does not fix the number of cosmological beginnings. Multiple source-free causal domains could be considered, but their number, joint interpretation, and observational implications would need additional assumptions. A later bounce, nucleation, or daughter universe inside existing actuality would have an antecedent actual context. It would therefore be a question about subsequent development, not by itself an example of genesis from absolute absence. Neither multiple Big Bangs nor their impossibility is deduced here.

10. Assumptions, results, and failure conditions

The contribution is a connected conditional framework. Its mathematical components include standard fermionic exclusion, finite birth processes, invariant quantum subsystems, elementary packing arguments, and exposure coupling. No priority is claimed for those methods. The proposed interpretation links a source-free origin conjecture to structural admission, stabilization, and explicit limits on retrospective inference.

The principal commitments and their consequences can be summarized as follows.

Component	Assumption or task	Result within this paper
Genesis	H3 admits positive actuality without antecedent actuality	A stated origin hypothesis, not an inevitability proof
Structural feedback	Relevant properties fix the represented response	Factorization criterion and explicit state-dependent admission
Finite quantum benchmark	Finite modes, no losses, positive gain, number-preserving ordinary response	Exact closure statistics with continuing internal dynamics
Packing benchmark	Bounded attribute range, positive exclusion scale, persistent accessibility	Finite capacity and almost-sure persistent closure
Residual additions	Bounded intensity, effect, amplification, and comparison error	A finite-horizon probability bound
Origin inference	Justified past response or surviving records	Conditional past restriction and an explicit matched-future limitation
Cosmological application	Physical variables, continuum behavior, conservation, and discriminating observations	A matching schema and unresolved physical obligations

Several failures are immediate. Infinite mode dimension removes the finite-capacity proof. Losses or number-changing ordinary response can reopen channels. Zero gain can leave admitted additions unrealized. A trivial internal generator produces no distinguishable closed-sector motion. Weakening accessibility can prevent maximum packing. Changing past response can defeat a finite-duration inference. Identical represented futures carry no likelihood distinction between the coupled origin histories.

Three tasks would materially advance the account. The origin argument needs independent support for H3, with its scope clearly specified. The structural models need physically motivated content rather than additional adjustable rules. A cosmological version needs a consequence tied to those independently motivated assumptions and a comparison with alternatives. Repeating successful mathematical examples alone would not complete those tasks.

11. Conclusion

The boundary hypothesis proposes that all physical actuality ultimately originates without antecedent actuality, and that properties and relations established with actuality constrain its further development. The origin claim remains a motivated conjecture. The constructions establish that structure-dependent admission, exact or operational stabilization, and continuing ordinary activity are mathematically compatible under explicit assumptions. They do not supply a mechanism acting on a preexisting state of nothingness.

The infinite-to-finite question has a precise conditional answer in the packing example: infinitely many candidate labels need not permit infinitely many compatible actual forms when a positive exclusion scale and bounded range are present. The origin of those structural properties is a separate question. The retrospective results also explain why a stable later universe would not by itself identify its origin.

A connection to a Big Bang remains a physical research objective. The cosmological matching conditions make that objective concrete without treating an abstract filling process as an already derived universe. The paper provides a self-contained framework in which the origin conjecture, its mathematical consequences, and the remaining physical questions can be evaluated separately and in relation to one another.

Appendix A. Operator and count calculations

A.1 An explicit finite representation

Label a basis of $F(\mathbb{C}^M)$ by binary masks b from zero to 2^M-1. Bit j records mode j. If bit j is unoccupied, $a_j^\dagger$ sends $|b\rangle$ to $|b+2^j\rangle$ multiplied by (-1) to the power of the number of occupied lower bits. If it is occupied, the result is zero. This defines finite matrices directly. Applying two distinct creation operators in opposite orders changes the sign once, and applying a creation twice gives zero. The annihilation matrices are their adjoints. Acting on each basis vector verifies the anticommutation relations, the number eigenvalue popcount(b), and (4).

For a number-sector vector, all $C_j^\dagger C_j$ are nonnegative. Their expectation sums to M-n. At n=M the zero sum makes every individual channel vanish. Tensoring with any internal space preserves these identities, including for states entangled between occupation and internal factors.

A.2 The adjoint gain identity

The adjoint dissipator applied to N is the sum of $C_j^\dagger N C_j$ minus one half of the anticommutator of $C_j^\dagger C_j$ and N, multiplied by κ. The relation $NC_j=C_j(N+I)$ gives $C_j^\dagger NC_j=C_j^\dagger C_j(N+I)$. Since $C_j^\dagger C_j$ commutes with N, the anticommutator term subtracts $C_j^\dagger C_j N$. The remaining term is $C_j^\dagger C_j$. Sum and use (4) to obtain κ(MI-N). The ordinary commutator contributes zero because [K,N]=0.

In the full sector, $C_{j\rho}=0$, $\rho C_j^\dagger=0$, and $C_j^\dagger C_j$ vanishes on both supports. The dissipator therefore annihilates every operator supported there, not only one selected pure state. This establishes the claimed protected internal response for arbitrary full-sector density matrices.

A.3 Forward equations and the harmonic identity

Write $p_j(t)$ for the probability of j additions from v vacancies and set $p_{-1}=0$. The forward equations are dot $p_j=\kappa(v-j+1)p_{j-1}-\kappa(v-j)p_j$, with $p_0(0)=1$ and all other initial probabilities zero. Differentiating (7) verifies every equation and normalization. The finite linear initial-value problem has a unique solution, so the probability law follows independently of any proposed physical identity of individual vacancies.

The mean closure time also equals the integral of one minus the cumulative distribution in (9). Expand that survival probability by the binomial theorem and integrate each exponential. In units κ=1 the result is the sum, from j=1 to v, of $(-1)^{j+1}$ binom(v,j)/j. To see that this equals H_v, integrate $[1-(1-x)^v]/x$ from zero to one. Expanding the numerator gives the alternating sum; writing the integrand as the finite geometric sum of $(1-x)^k$ for k from zero to v-1 gives H_v. Thus both derivations agree exactly.

Appendix B. Derivation of the biased packing formula

Let $m=n+1$ and let $z_1<...<z_m$ be a compatible inserted configuration. Put $y_i=2z_i-1$. Expanding the configuration and candidate weights in Section 6.2 produces a constant term, two terms linear in one bias, and a cross term. Reflection about 1/2 cancels the two linear terms.

For the cross term, decompose an m-configuration by deleting each possible inserted point z_i. For that deletion the product is y_i times the mean of the remaining n values. Summing over all deletions gives $[(\text{sum } y_i)^2-\text{sum } y_i^2]/n$. The insertion change of variables therefore gives the cross integral as $V_m/(nV_n)$ times the uniform expectation of that difference over Ω_m.

Set $R=1-n\delta>0$ and write $z_i=t_i+(i-1)\delta$, where the t_i are ordered uniform points in $[0,R]$. The sum of ordered points equals the sum of the original independent uniform points. Its variance is $mR^2/12$. Since $E[\text{sum } y_i]=0$, this gives $E[(\text{sum } y_i)^2]=mR^2/3$.

The density of t_i is proportional to $t^{i-1}(R-t)^{m-i}$. Normalizing by the beta integral gives $E[t_i]=Ri/(m+1)$ and $E[t_i^2]=R^2i(i+1)/((m+1)(m+2))$. Substituting into $y_i=2t_i+2(i-1)\delta-1$ and summing over i gives $E[\text{sum } y_i^2]=m(1+2n\delta^2)/3$.

Their difference is $-mn[2\delta-(n-2)\delta^2]/3$. Multiply by $V_m/(nV_n)$. Since $a_n=mV_m/V_n$, the cross integral is $-a_{nBn}$. Adding the constant term a_n gives (13). At maximum occupancy, acceptance is zero directly; no ratio on an empty configuration domain is needed.

This calculation assumes that ξ and χ are chosen before the current configuration and candidate draws. If the candidate bias is selected after inspecting the freshly sampled configuration, this particular factorization need not hold. The broader lower-bound theorem remains available when its stated accessibility assumptions hold.

An independent way to check the integral is to expand the two sampling weights into monomials in ordered unit-interval coordinates $u_1<...<u_m$. For nonnegative integer exponents e_i, successive integration gives their normalized monomial expectation as m! divided by the product, over k from one to m, of $k+e_1+...+e_k$. Appendix D applies this rule to every deletion term with exact rational arithmetic, without substituting the closed expression for B_n.

Appendix C. Verification scope and numerical record

The eleven mathematical propositions have written proofs; the origin claims H1-H4 retain the status specified in Section 2. The program printed in Appendix D performs additional finite checks. It uses Python 3.10 or later and NumPy, with no dependencies on other project files. The complete visible listing is sufficient to inspect and reproduce the checks without a separate code attachment.

The checks cover the finite creation algebra and adjoint gain identities, full-sector dissipation, count normalization and forward equations, exact harmonic-mean identities, the geometric bias fractions, transition composition, and matched-future birth times. The packing check independently integrates the polynomial sampling weights over ordered simplices using rational arithmetic and compares the result with (13). These checks test implementations and selected finite cases. They do not establish H3, mathematical originality, or empirical agreement with cosmology.

The reported record for this draft is generated by executing that exact program. Its compact summary is reproduced below. Numerical tolerances and case counts are specified in the listing, so the record can be regenerated without earlier manuscripts, scripts, or correspondence.

Check	Cases or result
Finite operator representations	5
Count probability and forward-equation cases	63
Maximum forward-equation residual	2.599e-13
Exact harmonic identities	16
Exact independent packing integrals	70
Transition-composition cases	72
Coupled birth thresholds	20,001
Biased mean opportunity count	694.088582686
Program assertions	All passed

Appendix D. Complete verification program

The following program prints its summary as JSON. All variables refer to the represented positive models. The count and transition tests use dimensionless accumulated exposure. The matched-history test separately uses $\kappa_0=0.7$ and $\tau=1.3$ in consistent units. Neither convention assigns time or units to the proposed absence boundary.

```
"""Finite checks accompanying the boundary hypothesis manuscript."""
import json
import math
from fractions import Fraction as F
import numpy as np


def creations(modes):
    size = 1 << modes
    result = []
    for j in range(modes):
        matrix = np.zeros((size, size), dtype=np.int64)
        for mask in range(size):
            if not mask & (1 << j):
                lower = (mask & ((1 << j) - 1)).bit_count()
                matrix[mask | (1 << j), mask] = (-1) ** lower
        result.append(matrix)
    return result


def count_law(vacancies, time):
    unfilled = math.exp(-time)
    filled = -math.expm1(-time)
    return np.array([
        math.comb(vacancies, j) * filled ** j
        * unfilled ** (vacancies - j)
        for j in range(vacancies + 1)
    ])


def transitions(capacity, exposure):
    result = np.zeros((capacity, capacity))
    for initial in range(1, capacity + 1):
        result[initial - 1:, initial - 1] = count_law(
            capacity - initial, exposure)
    return result


def ordered_moment(powers):
    """Normalized monomial integral on 0 < u_1 < ... < u_m < 1."""
    value = F(math.factorial(len(powers)))
    degree = 0
    for index, power in enumerate(powers, 1):
        degree += power
        value /= index + degree
    return value


def packing_integral(n, delta, xi, chi):
    """Integrate each possible insertion's two sampling weights exactly."""
    m = n + 1
    width = 1 - n * delta
    if width <= 0:
        return F(0)
    volume_ratio = (width ** m / math.factorial(m)) / (
        (1 - (n - 1) * delta) ** n / math.factorial(n))
    constants = [2 * i * delta - 1 for i in range(m)]
    first = []
    for i in range(m):
        powers = [0] * m
```

```
            powers[i] = 1
            first.append(ordered_moment(powers))
        means = [c + 2 * width * u for c, u in zip(constants, first)]
        integrated = F(0)
        for inserted in range(m):
            others = [j for j in range(m) if j != inserted]
            cross = F(0)
            for j in others:
                powers = [0] * m
                powers[inserted] = powers[j] = 1
                cross += (constants[inserted] * constants[j]
                          + 2 * width * (constants[inserted] * first[j]
                                         + constants[j] * first[inserted])
                          + 4 * width ** 2 * ordered_moment(powers))
            integrated += (1 + chi * means[inserted]
                           + xi * sum(means[j] for j in others) / n
                           + xi * chi * cross / n)
        return volume_ratio * integrated


    def run():
        report = {}
        operator_cases = 0
        for modes in range(1, 6):
            channels = creations(modes)
            size = 1 << modes
            identity = np.eye(size, dtype=np.int64)
            number = np.diag([b.bit_count() for b in range(size)])
            available = np.zeros_like(identity)
            adjoint = np.zeros_like(identity)
            for j, creation in enumerate(channels):
                annihilation = creation.T
                for k, other in enumerate(channels):
                    target = identity if j == k else 0 * identity
                    assert np.array_equal(
                        annihilation @ other + other @ annihilation,
                        target)
                    assert not np.any(creation @ other + other @ creation)
                product = annihilation @ creation
                available += product
                adjoint += (2 * annihilation @ number @ creation
                            - product @ number - number @ product)
            assert np.array_equal(available, modes * identity - number)
            assert np.array_equal(adjoint, 2 * available)
            full = np.zeros((size, size), dtype=np.int64)
            full[-1, -1] = 1
            for creation in channels:
                product = creation.T @ creation
                dissipator = (2 * creation @ full @ creation.T
                              - product @ full - full @ product)
                assert not np.any(dissipator)
            operator_cases += 1
        report['operator_cases'] = operator_cases

        count_cases = 0
        max_forward_error = 0.0
        for vacancies in range(9):
            generator = np.zeros((vacancies + 1, vacancies + 1))
            for j in range(vacancies):
                generator[j, j] = -(vacancies - j)
                generator[j + 1, j] = vacancies - j
            for time in (0.0, 0.01, 0.1, 1.0, 2.0, 5.0, 20.0):
                probabilities = count_law(vacancies, time)
                derivative = np.array([
                    math.comb(vacancies, j) * sum(
                        (-1) ** k * math.comb(j, k)
                        * -(vacancies - j + k)
                        * math.exp(-time * (vacancies - j + k))
```

```
                for k in range(j + 1))
            for j in range(vacancies + 1)
        ])
        error = np.max(np.abs(
            derivative - generator @ probabilities))
        max_forward_error = max(max_forward_error, float(error))
        assert error < 2e-9
        assert abs(probabilities.sum() - 1) < 2e-12
        assert np.all(probabilities >= 0)
        expected = vacancies * -math.expm1(-time)
        assert abs(probabilities @ np.arange(vacancies + 1)
                   - expected) < 2e-12
        expected_variance = expected * math.exp(-time)
        variance = probabilities @ (
            np.arange(vacancies + 1) - expected) ** 2
        assert abs(variance - expected_variance) < 2e-12
        assert abs(probabilities[-1]
                   - (-math.expm1(-time)) ** vacancies) < 2e-12
        count_cases += 1
report['count_cases'] = count_cases
report['max_forward_error'] = max_forward_error

for v in range(1, 17):
    integral = sum((F((-1) ** (j + 1) * math.comb(v, j), j)
                    for j in range(1, v + 1)), F(0))
    harmonic = sum((F(1, j) for j in range(1, v + 1)), F(0))
    assert integral == harmonic
report['exact_harmonic_cases'] = 16

delta, xi, chi = F(3, 10), F(1, 2), F(3, 4)
expected = [F(35819, 80000), F(148, 1225), F(749, 512000)]
biased = []
for n in range(1, 4):
    fraction = (1 - n * delta) ** (n + 1)
    fraction /= (1 - (n - 1) * delta) ** n
    bias = (2 * delta - (n - 2) * delta ** 2) / 3
    biased.append(fraction * (1 - xi * chi * bias))
assert biased == expected
report['packing_fractions'] = [str(x) for x in biased]
report['mean_opportunities'] = float(sum(F(1) / x for x in biased))

packing_cases = 0
for gap in (F(2, 3), F(1, 2), F(1, 3), F(3, 10), F(1, 4), F(1, 5)):
    capacity = math.ceil(1 / gap)
    for n in range(1, capacity):
        uniform = (1 - n * gap) ** (n + 1)
        uniform /= (1 - (n - 1) * gap) ** n
        coefficient = (2 * gap - (n - 2) * gap ** 2) / 3
        for x_bias, c_bias in ((F(0), F(0)), (F(1, 2), F(3, 4)),
                               (F(-1, 2), F(3, 4)),
                               (F(9, 10), F(-9, 10)),
                               (F(9, 10), F(9, 10))):
            direct = packing_integral(n, gap, x_bias, c_bias)
            formula = uniform * (1 - x_bias * c_bias * coefficient)
            assert direct == formula
            assert 0 < direct <= 1
            packing_cases += 1
    assert packing_integral(capacity, gap, F(1, 2), F(3, 4)) == 0
report['packing_integral_cases'] = packing_cases

composition_cases = 0
for capacity in range(1, 13):
    for first, second in ((0, 0), (0, 1), (0.1, 0.4),
                          (1, 2), (4, 0.05), (5, 7)):
        direct = transitions(capacity, first + second)
        composed = (transitions(capacity, second)
                    @ transitions(capacity, first))
        assert np.max(np.abs(direct - composed)) < 3e-12
```

```
            composition_cases += 1
    report['composition_cases'] = composition_cases

    thresholds = np.linspace(0.0001, 8.0, 20001)
    kappa, tau = 0.7, 1.3
    exposure = kappa * tau
    r_birth = -tau + thresholds / kappa
    b_birth = np.where(thresholds < exposure,
                       tau * np.log(thresholds / exposure),
                       (thresholds - exposure) / kappa)
    future = thresholds >= exposure
    assert np.max(np.abs(r_birth[future] - b_birth[future])) < 3e-12
    for time in (0, 0.2, 1, 3, 10):
        assert np.array_equal(r_birth <= time, b_birth <= time)
    report['matched_future_thresholds'] = len(thresholds)
    report['checks_passed'] = True
    return report


if __name__ == '__main__':
    print(json.dumps(run(), indent=2))
```

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